• **ENGINE—SPARK KNOCK—SPARK KNOCK ON STEADY CRUISE OR LIGHT ACCELERATION—VEHICLES EQUIPPED WITH 2.5L ENGINE** • **NOISE—SPARK KNOCK—SPARK KNOCK ON STEADY CRUISE OR LIGHT ACCELERATION—VEHICLES EQUIPPED WITH 2.5L ENGINE**	**Article No. 01-1-4**

FORD: 1998-2000 CONTOUR

MERCURY: 1998-2000 MYSTIQUE
1999-2000 COUGAR

This article applies to 1998-2000 Contour/Mystique vehicles equipped with 2.5L engine built from 2/2/1998 only and all 1999-2000 Cougars equipped with 2.5L engine.

ISSUE

A spark knock or pre-ignition condition may exist on some vehicles with an automatic or manual transmission during steady cruise driving or under light acceleration.

ACTION

Reprogram the Powertrain Control Module (PCM). The revised calibration will reduce the potential for spark knock. Refer to the following Service Procedure for details.

NOTE

SOME LIGHT DETONATION ON TIP-IN IS NORMAL AND CONSIDERED CHARACTERISTIC.

SERVICE PROCEDURE

1. Verify spark knock condition.
2. Perform the diagnostic self-test to check for Diagnostic Trouble Codes (DTCs). Resolve all codes that are present using the appropriate Powertrain Control/Emissions Diagnosis (PC/ED) Service Manual before proceeding.
3. Using a New Generation Star (NGS) Tester or the Worldwide Diagnostic System (WDS) only (this function is not available on Service Bay Diagnostic System (SBDS), access the PCM identification function to obtain the part number of the PCM.
4. Use the PCM part number to identify the vehicle's existing calibration by referring to the following PCM Calibration Information Chart.

NOTE

IF THE PCM PART NUMBER OBTAINED HAS SEVEN (7) DIGITS, DISREGARD THE LAST (7TH) DIGIT AND USE ONLY THE FIRST SIX (6) DIGITS WHEN REFERRING TO THE CHART. IF THE PCM PART NUMBER OBTAINED HAS SIX (6) DIGITS, DISREGARD THE LAST (6TH) DIGIT AND USE ONLY THE FIRST FIVE (5) DIGITS WHEN REFERRING TO THE CHART.

5. Once the existing calibration has been identified, reprogram the PCM to the latest calibration revision level.

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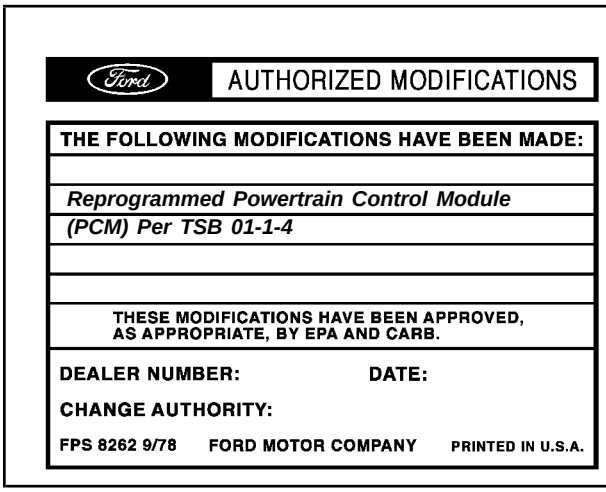
PCM CALIBRATION INFORMATION							
Application	Old Part Number (-12A650-)	Old Tear Tag	New Part Number (-12A650-)	New Tear Tag	Old Calibration	New Calibration	NGS/WDS Qualifier
Cougar CD4E/50S	XS8A-MC	DMX1	XS8Z-MD	DMX2	9LCA-BWG	9LCA-BWG	9LCA-BW Calibration
Cougar MTX/50S	XS8A-PD	ZKZx	XS8Z-PE	ZKZx	9LCM-BWG	9LCM-BWG	9LCM-BW Calibration
Contour/Mystique, Cougar, CD4E/50S	XS2A-XC	VNMx	XS2Z-XD	VNMx	9LCA-BVG	9LCA-BVH	9LCA-BV Calibration
Contour/Mystique CD4E/Mex	XS2A-YC	TRMO	XS2Z-YD	TRMO	9LCA-BVGME	9LCA-BVHME	...
Contour/Mystique, Cougar MTX/50S	XS2A-ZC	GGBx	XS2Z-ZD	GGBx	9LCM-BVG	9LCM-BVH	9LCM-BV Calibration
Contour/Mystique MTX/Mex	XS2A-ABC	HKG	XS2Z-ABD	HKG	9LCM-BVGME	9LCM-BVHME	...
Contour/Mystique, Cougar CD4E/Calif	XS2A-HF	MZZx	XS2Z-HG	MZZx	9LCA-BCD	9LCA-BCD	9LCA-BC Calibration
Contour/Mystique, Cougar CD4E/49S	XS2A-GF	LXPx	XS2Z-GG	LXPx	9LCA-ACD	9LCA-ACD	9LCA-AC Calibration
Contour/Mystique CD4E/Mex	XS2A-VF	PPC	XS2Z-VG	PPC	9LCA-ACDME	9LCA-ACDME	...
Contour/Mystique, Cougar MTX/Calif	XS2A-VG	DARx	XS2Z-FH	DARx	9LCM-BCE	9LCM-BCE	9LCM-BC Calibration
Contour/Mystique, Cougar MTX/49S	XS2A-EG	JTA	XS2Z-EH	JTA	9LCM-ACE	9LCM-ACE	9LCM-AC Calibration
Contour/Mystique MTX/Mex	XS2A-PG	NRE	XS2Z-PH	NRE	9LCM-ACEME	9LCM-ACEME	...
Cougar, CD4E/Calif	XS8A-HF	HYAx	XS8Z-HG	HYAx	9LCA-BDD	9LCA-BDD	9LCA-BD Calibration
Cougar, CD4E/49S	XS8A-GF	GGRx	XS8Z-GG	GGRx	9LCA-ADD	9LCA-ADD	9LCA-AD Calibration
Cougar, MTX/Calif	XS8A-FG	FWFx	XS8Z-FH	FWFx	9LCM-BDE	9LCM-BDE	9LCM-BD Calibration
Cougar, MTX/49S	XS8A-EG	EDLx	XS8Z-EH	EDLx	9LCM-ADE	9LCM-ADE	9LCM-AD Calibration
Contour/Mystique, Cougar, CD4E, CUSH/Calif	98BB-CAD	BLCx	XS2Z-CAE	BLCx	9LCA-BSH	9LCA-BSH	9LCA-BA trans shift TSB 98-13-11
Contour/Mystique, Cougar CD4E/Calif	98BB-AMG	PLAx	XS8Z-AMH	PLAx	9LCA-BAE	9LCA-BAE	9LCM-AD Calibration
Contour/Mystique, Cougar, CD4E, CUSH/49S	98BB-BZC	AZBx	XS2Z-BZD	AZBx	9LCA-ASG	9LCA-ASG	9LCA-AA trans shift TSB 98-13-11
Contour/Mystique, Cougar CD4E/49S	98BB-ALG	MJJx	XS8Z-ALH	MJJx	9LCA-AAE	9LCA-AAE	9LCA-AA Calibration
Contour/Mystique, CD4E, CUSH/Mex	98BB-CBC	CDA	XS2Z-CBD	CDA	9LCA-ASGME	9LCA-ASGME	...
Contour/Mystique, CD4E, Mex	98BB-ANG	RRQ	XS2Z-ANH	RRQ	9LCA-AAEME	9LCA-AAEME	...
Contour/Mystique, Cougar MTX/Calif	98BB-AKG	LTZx	XS2Z-AKH	LTZx	9LCM-BAF	9LCM-BAF	9LCM-BA Calibration
Contour/Mystique, Cougar MTX/49S	98BB-AJG	KKTx	XS2Z-AJH	KKTx	9LCM-AAF	9LCM-AAF	9LCM-AA Calibration

Application	Old Part Number (-12A650-)	Old Tear Tag	New Part Number (-12A650-)	New Tear Tag	Old Calibration	New Calibration	NGS/WDS Qualifier
Contour/Mystique, MTX, Mex	98BB-AZG	JGJx	XS2Z-AZH	JGJx	9LCM-AAFME	9LCM-AAFME	...
Cougar, CD4E, CUSH/Calif	98BB-CFC	YEBx	XS8Z-CFD	YEBx	9LCA-BTH	9LCA-BTH	9LCA-BB trans shift TSB 98-13-11
Cougar, CD4E/Calif	98BB-ATG	XPPx	XS8Z-ATH	XPPx	9LCA-BBE	9LCA-BBE	9LCA-BB Calibration
Cougar, CD4E, CUSH/49S	98BB-CEC	FPT	XS8Z-CED	FPT	9LCA-ATG	9LCA-ATG	9LCA-AB trans shift TSB 98-13-11
Cougar, CD4E/49S	98BB-ASG	VLMx	XS8Z-ASH	VLMx	9LCA-ABE	9LCA-ABE	9LCA-AB Calibration
Cougar, MTX/Calif	98BB-ARG	TRWx	XS8Z-ARH	TRWx	9LCM-BBF	9LCM-BBF	9LCM-BB Calibration
Cougar, MTX/49S	98BB-APG	SSDx	XS8Z-APH	SSDx	9LCM-ABF	9LCM-ABF	9LCM-AB Calibration

Obtain an Authorized Modifications Decal (FPS 8262 - obtainable through DOES II, 25/pkg) and list the date, dealer number, and summary of modifications performed. Select a prominent place adjacent to the Vehicle Emission Control Information Decal suitable for installing the Authorized Modifications Decal. Clean the area, install the decal, and cover it with a clear plastic decal shield.



OTHER APPLICABLE ARTICLES: NONE

WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage And Emissions Warranty Coverage

OPERATION	DESCRIPTION	TIME
010104A	Check And Reprogram Powertrain Control Module (PCM)	0.6 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
12A650	55

OASIS CODES: 497000, 615000, 615500, 615600, 702100